

Installation and Operational Qualification Testing For R Statistical Software

R version 4.6.1 (2026-06-24 ucrt)

Architecture: x86_64

Platform: x86_64-w64-mingw32

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Introduction

The R software being tested in this report was downloaded from The Comprehensive R Archive Network (CRAN):

<http://cran.r-project.org/>

or a CRAN mirror:

<http://cran.us.r-project.org/mirrors.html>

The R software was installed on this computer in a manner consistent with “The R Installation and Administration Manual” (the Manual) which is available from:

<http://cran.r-project.org/doc/manuals/R-admin.html>

The Manual provides recommendations for post-installation testing of R. The procedures for this testing are described in:

http://cran.r-project.org/doc/manuals/R-admin.html#Testing-a-Unix_002dalike-Installation

for Unix, Linux and OSX installations and:

<http://cran.r-project.org/doc/manuals/R-admin.html#Testing-a-Windows-Installation>

for Windows installations. The results contained within this report are based upon an automated implementation of the procedures described in the Manual.

General guidance for the use of R in regulated clinical trials has been provided by the R Foundation in a document entitled:

R: Regulatory Compliance and Validation Issues

A Guidance Document for the Use of R in Regulated Clinical Trial Environments

which is available from:

<http://www.r-project.org/doc/R-FDA.pdf>

The above document describes various characteristics of R, including the Software Development Life Cycle (SDLC) and relevant aspects of 21 CFR Part 11 compliance issues as they may pertain to the use of R for statistical analysis applications for clinical trials.

The output on the following pages of this report describe various technical characteristics of the computer upon which R is running, the R installation, current R session information and is followed by a series of tests for the so-called “Base” and “Recommended” packages which are a part of the official R distribution, as released by the R Foundation. Any installed packages that are not “Base” or “Recommended” packages are not part of this validation procedure and are ignored.

Installation Qualification (IQ)

The following is the output of `R.home()`, showing where R was installed on this computer:

```
1 C:/PROGRA~1/R/R-46~1.1
```

The following is the output of `system("R -e 'q()'")`, presenting the R welcome banner as displayed from a default R console (terminal) to show the R console correctly running and then exiting:

```
1
2 R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
3 Copyright (C) 2026 The R Foundation for Statistical Computing
4 Platform: x86_64-w64-mingw32/x64
5
6 R is free software and comes with ABSOLUTELY NO WARRANTY.
7 You are welcome to redistribute it under certain conditions.
8 Type 'license()' or 'licence()' for distribution details.
9
10 Natural language support but running in an English locale
11
12 R is a collaborative project with many contributors.
13 Type 'contributors()' for more information and
14 'citation()' on how to cite R or R packages in publications.
15
16 Type 'demo()' for some demos, 'help()' for on-line help, or
17 'help.start()' for an HTML browser interface to help.
18 Type 'q()' to quit R.
19
20 > q()
```

The following is the output of `Sys.info()`, defining some details about the current system upon which R is running and user information:

1	sysname	release	version	nodename	machine
2	"Windows"	"10 x64"	"build 26200"	"SAMIRA-LENOVO"	"x86_64"
3	login	user	effective_user	udomain	
4	"samir"	"samir"	"samir"	"SAMIRA-LENOVO"	

The following is the output of `.Platform`, defining some details of the platform upon which R was built (compiled):

```
1 $OS.type
2 [1] "windows"
3
4 $file.sep
5 [1] "/"
6
7 $dynlib.ext
8 [1] ".dll"
9
10 $GUI
11 [1] "RTerm"
12
13 $endian
14 [1] "little"
15
16 $pkgType
17 [1] "win.binary"
18
19 $path.sep
20 [1] ";"
21
22 $r_arch
23 [1] "x64"
```

The following is the output of `R.version`, defining detailed information on the currently running version of R:

```
1
2 platform      -
3 arch          x86_64-w64-mingw32
4 os            mingw32
5 crt            ucrt
6 system        x86_64, mingw32
7 status
8 major         4
9 minor         6.1
10 year          2026
11 month         06
12 day           24
13 svn rev       90187
14 language      R
15 version.string R version 4.6.1 (2026-06-24 ucrt)
16 nickname      Happy Hop
```

The following is the output of `.Machine`, defining the numerical characteristics of the computer upon which R is running:

```
1 $double.eps
2 [1] 2.220446e-16
3
4 $double.neg.eps
5 [1] 1.110223e-16
6
7 $double.xmin
8 [1] 2.225074e-308
9
10 $double.xmax
11 [1] 1.797693e+308
12
13 $double.base
14 [1] 2
15
16 $double.digits
17 [1] 53
18
19 $double.rounding
20 [1] 5
21
22 $double.guard
23 [1] 0
24
25 $double.ulp.digits
26 [1] -52
27
28 $double.neg.ulp.digits
29 [1] -53
30
31 $double.exponent
32 [1] 11
33
34 $double.min.exp
35 [1] -1022
36
37 $double.max.exp
38 [1] 1024
39
40 $integer.max
41 [1] 2147483647
42
43 $sizeof.long
44 [1] 4
45
46 $sizeof.longlong
47 [1] 8
48
49 $sizeof.longdouble
50 [1] 16
51
52 $sizeof.pointer
53 [1] 8
54
55 $sizeof.time_t
56 [1] 8
57
58 $longdouble.eps
59 [1] 1.084202e-19
60
61 $longdouble.neg.eps
62 [1] 5.421011e-20
63
64 $longdouble.digits
65 [1] 64
66
67 $longdouble.rounding
68 [1] 5
69
70 $longdouble.guard
71 [1] 0
72
73 $longdouble.ulp.digits
74 [1] -63
75
76 $longdouble.neg.ulp.digits
77 [1] -64
78
79 $longdouble.exponent
```

```
80 | [1] 15
81 |
82 | $longdouble.min.exp
83 | [1] -16382
84 |
85 | $longdouble.max.exp
86 | [1] 16384
```

The following is the output of `sessionInfo()`, defining current R version, locale information and attached packages:

```
1 R version 4.6.1 (2026-06-24 ucrt)
2 Platform: x86_64-w64-mingw32/x64
3 Running under: Windows 11 x64 (build 26200)
4
5 Matrix products: default
6   LAPACK version 3.12.1
7
8 locale:
9 [1] LC_COLLATE=German_Switzerland.utf8 LC_CTYPE=German_Switzerland.utf8
10 [3] LC_MONETARY=German_Switzerland.utf8 LC_NUMERIC=C
11 [5] LC_TIME=German_Switzerland.utf8
12
13 time zone: Europe/Zurich
14 tzcode source: internal
15
16 attached base packages:
17 [1] stats      graphics  grDevices  utils      datasets  methods    base
18
19 loaded via a namespace (and not attached):
20 [1] compiler_4.6.1
```

The following is the output of `.libPaths()`, the current package library location; may be more than one folder:

```
1 [1] "C:/Users/samir/AppData/Local/R/win-library/4.6"
2 [2] "C:/Program Files/R/R-4.6.1/library"
```

The following is the output of `rmarkdown::pandoc_version()` listing the version of Pandoc used to render the report:

```
1 [1] '3.6.3'
```

The following is the output of `tinytex::tlmgr_version()` listing the version and installation path of TinyTeX used to render the report to pdf:

```
1 tlmgr revision 79639 (2026-07-10 18:45:34 +0200)
2 tlmgr using installation: C:/Users/samir/AppData/Roaming/TinyTeX
3 TeX Live (https://tug.org/texlive) version 2026
```

R Core Operational Qualification - System Tests (OQ)

The following is the output of `testInstalledBasic("both")`, which runs a series of core system-wide operational tests of the R installation, including various regression tests:

```
1
2 R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
3 Copyright (C) 2026 The R Foundation for Statistical Computing
4 Platform: x86_64-w64-mingw32/x64
5
6 R is free software and comes with ABSOLUTELY NO WARRANTY.
7 You are welcome to redistribute it under certain conditions.
8 Type 'license()' or 'licence()' for distribution details.
9
10 Natural language support but running in an English locale
11
12 R is a collaborative project with many contributors.
13 Type 'contributors()' for more information and
14 'citation()' on how to cite R or R packages in publications.
15
16 Type 'demo()' for some demos, 'help()' for on-line help, or
17 'help.start()' for an HTML browser interface to help.
18 Type 'q()' to quit R.
19
20 >
21 > options(echo = FALSE)
22 running strict specific tests
23   running code in 'eval-etc.R'
24   comparing 'eval-etc.Rout' to 'eval-etc.Rout.save' ... OK
25   running code in 'simple-true.R'
26   comparing 'simple-true.Rout' to 'simple-true.Rout.save' ... OK
27   running code in 'arith-true.R'
28   comparing 'arith-true.Rout' to 'arith-true.Rout.save' ... OK
29   running code in 'lm-tests.R'
30   comparing 'lm-tests.Rout' to 'lm-tests.Rout.save' ... OK
31   running code in 'ok-errors.R'
32   comparing 'ok-errors.Rout' to 'ok-errors.Rout.save' ... OK
33   running code in 'method-dispatch.R'
34   comparing 'method-dispatch.Rout' to 'method-dispatch.Rout.save' ... OK
35   running code in 'array-subset.R'
36   running code in 'p-r-random-tests.R'
37   comparing 'p-r-random-tests.Rout' to 'p-r-random-tests.Rout.save' ... OK
38   running code in 'd-p-q-r-tst-2.R'
39   running code in 'any-all.R'
40   comparing 'any-all.Rout' to 'any-all.Rout.save' ... OK
41   running code in 'structure.R'
42   comparing 'structure.Rout' to 'structure.Rout.save' ... OK
43   running code in 'd-p-q-r-tests.R'
44   comparing 'd-p-q-r-tests.Rout' to 'd-p-q-r-tests.Rout.save' ... OK
45 running sloppy specific tests
46   running code in 'complex.R'
47   comparing 'complex.Rout' to 'complex.Rout.save' ... OK
48   running code in 'print-tests.R'
49   comparing 'print-tests.Rout' to 'print-tests.Rout.save' ... OK
50   running code in 'lapack.R'
51   comparing 'lapack.Rout' to 'lapack.Rout.save' ... OK
52   running code in 'datasets.R'
53   comparing 'datasets.Rout' to 'datasets.Rout.save' ... OK
54   running code in 'datetime.R'
55   comparing 'datetime.Rout' to 'datetime.Rout.save' ... OK
56   running code in 'ie60559.R'
57   comparing 'ie60559.Rout' to 'ie60559.Rout.save' ... OK
58 running regression tests
59   running code in 'reg-tests-1a.R'
60   running code in 'reg-tests-1b.R'
61   running code in 'reg-tests-1c.R'
62   running code in 'reg-tests-1d.R'
63   running code in 'reg-tests-1e.R'
64   running code in 'reg-tests-2.R'
65   comparing 'reg-tests-2.Rout' to 'reg-tests-2.Rout.save' ... OK
66   running code in 'reg-examples1.R'
67   running code in 'reg-examples2.R'
68   running code in 'reg-packages.R'
69   running code in 'reg-S4-examples.R'
70   running code in 'classes-methods.R'
71   running code in 'datetime3.R'
72   running code in 'p-qbata-strict-tst.R'
73   running code in 'reg-IO.R'
74   comparing 'reg-IO.Rout' to 'reg-IO.Rout.save' ... OK
```

```
75 running code in 'reg-I02.R'
76 comparing 'reg-I02.Rout' to 'reg-I02.Rout.save' ... OK
77 running code in 'reg-plot.R'
78 comparing 'reg-plot.pdf' to 'reg-plot.pdf.save' ...OK
79 running code in 'reg-S4.R'
80 comparing 'reg-S4.Rout' to 'reg-S4.Rout.save' ... OK
81 running code in 'reg-BLAS.R'
82 running code in 'reg-encodings.R'
83 running code in 'reg-translation.R'
84 running code in 'reg-tests-3.R'
85 comparing 'reg-tests-3.Rout' to 'reg-tests-3.Rout.save' ... OK
86 running code in 'reg-examples3.R'
87 comparing 'reg-examples3.Rout' to 'reg-examples3.Rout.save' ... OK
88 running tests of plotting Latin-1
89 expect failure or some differences if not in a Latin or UTF-8 locale
90 running code in 'reg-plot-latin1.R'
91 comparing 'reg-plot-latin1.pdf' to 'reg-plot-latin1.pdf.save' ...OK
92
93
94 Test suite result: PASS
```

The final line of the above output displays the status of running the above tests. **PASS** indicates a successful running of the tests, a **FAIL** would indicate that an error was detected during the running of the tests.

There may be some tests where the result of performing a **diff** on two files that were being compared demonstrate a content difference that may or may not be relevant and may be dependent upon locale settings. Any such differences displayed in the above output should be reviewed in detail to determine their relevance to the Operational Qualification of this R installation.

R Base Package Operational Qualification - Package Examples (OQ)

The following is the output of `testInstalledPackages(outDir = "IQ-OQ-TestOutput", scope = "base", types = "examples", errorsAreFatal = FALSE)`, which runs a series of operational tests of the R Base package code examples:

```
1
2 R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
3 Copyright (C) 2026 The R Foundation for Statistical Computing
4 Platform: x86_64-w64-mingw32/x64
5
6 R is free software and comes with ABSOLUTELY NO WARRANTY.
7 You are welcome to redistribute it under certain conditions.
8 Type 'license()' or 'licence()' for distribution details.
9
10 Natural language support but running in an English locale
11
12 R is a collaborative project with many contributors.
13 Type 'contributors()' for more information and
14 'citation()' on how to cite R or R packages in publications.
15
16 Type 'demo()' for some demos, 'help()' for on-line help, or
17 'help.start()' for an HTML browser interface to help.
18 Type 'q()' to quit R.
19
20 >
21 > options(echo = FALSE)
22 Testing examples for package 'base'
23 Testing examples for package 'tools'
24   comparing 'tools-Ex.Rout' to 'tools-Ex.Rout.save' ... OK
25 Testing examples for package 'utils'
26 Testing examples for package 'grDevices'
27   comparing 'grDevices-Ex.Rout' to 'grDevices-Ex.Rout.save' ... OK
28 Testing examples for package 'graphics'
29   comparing 'graphics-Ex.Rout' to 'graphics-Ex.Rout.save' ... OK
30 Testing examples for package 'stats'
31   comparing 'stats-Ex.Rout' to 'stats-Ex.Rout.save' ... OK
32 Testing examples for package 'datasets'
33   comparing 'datasets-Ex.Rout' to 'datasets-Ex.Rout.save' ... OK
34 Testing examples for package 'methods'
35 Testing examples for package 'grid'
36   comparing 'grid-Ex.Rout' to 'grid-Ex.Rout.save' ... OK
37 Testing examples for package 'splines'
38   comparing 'splines-Ex.Rout' to 'splines-Ex.Rout.save' ... OK
39 Testing examples for package 'stats4'
40   comparing 'stats4-Ex.Rout' to 'stats4-Ex.Rout.save' ... OK
41 Testing examples for package 'tcltk'
42 Testing examples for package 'compiler'
43 Testing examples for package 'parallel'
44
45
46 Test suite result: PASS
```

The final line of the above output displays the status of running the above tests. **PASS** indicates a successful running of the tests, a **FAIL** would indicate that an error was detected during the running of the tests.

There may be some tests where the result of performing a **diff** on two files that were being compared demonstrate a content difference that may or may not be relevant and may be dependent upon locale settings. Any such differences displayed in the above output should be reviewed in detail to determine their relevance to the Operational Qualification of this R installation.

R Base Package Operational Qualification - Package Vignettes (OQ)

The following is the output of `testInstalledPackages(outDir = "IQ-OQ-TestOutput", scope = "base", types = "vignettes", errorsAreFatal = FALSE)`, which runs a series of operational tests of the R Base package vignette code examples:

```
1
2 R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
3 Copyright (C) 2026 The R Foundation for Statistical Computing
4 Platform: x86_64-w64-mingw32/x64
5
6 R is free software and comes with ABSOLUTELY NO WARRANTY.
7 You are welcome to redistribute it under certain conditions.
8 Type 'license()' or 'licence()' for distribution details.
9
10 Natural language support but running in an English locale
11
12 R is a collaborative project with many contributors.
13 Type 'contributors()' for more information and
14 'citation()' on how to cite R or R packages in publications.
15
16 Type 'demo()' for some demos, 'help()' for on-line help, or
17 'help.start()' for an HTML browser interface to help.
18 Type 'q()' to quit R.
19
20 >
21 > options(echo = FALSE)
22 Running vignettes for package 'utils'
23   Running 'Sweave.Rnw'
24 Running vignettes for package 'stats'
25   Running 'reshape.Rnw'
26 Running vignettes for package 'grid'
27   Running 'displaylist.Rnw'
28   Running 'frame.Rnw'
29   Running 'grid.Rnw'
30   Running 'grobs.Rnw'
31   Running 'interactive.Rnw'
32   Running 'locndimn.Rnw'
33   Running 'moveline.Rnw'
34   Running 'nonfinite.Rnw'
35   Running 'plotexample.Rnw'
36   Running 'rotated.Rnw'
37   Running 'saveload.Rnw'
38   Running 'sharing.Rnw'
39   Running 'viewports.Rnw'
40 Running vignettes for package 'parallel'
41   Running 'parallel.Rnw'
42
43
44 Test suite result: PASS
```

The final line of the above output displays the status of running the above tests. **PASS** indicates a successful running of the tests, a **FAIL** would indicate that an error was detected during the running of the tests.

There may be some tests where the result of performing a `diff` on two files that were being compared demonstrate a content difference that may or may not be relevant and may be dependent upon locale settings. Any such differences displayed in the above output should be reviewed in detail to determine their relevance to the Operational Qualification of this R installation.

R Recommended Package Operational Qualification - Package Examples (OQ)

The following is the output of `testInstalledPackages(outDir = "IQ-OQ-TestOutput", scope = "recommended", types = "examples", errorsAreFatal = FALSE)`, which runs a series of operational tests of the R Recommended package code examples:

```
1
2 R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
3 Copyright (C) 2026 The R Foundation for Statistical Computing
4 Platform: x86_64-w64-mingw32/x64
5
6 R is free software and comes with ABSOLUTELY NO WARRANTY.
7 You are welcome to redistribute it under certain conditions.
8 Type 'license()' or 'licence()' for distribution details.
9
10 Natural language support but running in an English locale
11
12 R is a collaborative project with many contributors.
13 Type 'contributors()' for more information and
14 'citation()' on how to cite R or R packages in publications.
15
16 Type 'demo()' for some demos, 'help()' for on-line help, or
17 'help.start()' for an HTML browser interface to help.
18 Type 'q()' to quit R.
19
20 >
21 > options(echo = FALSE)
22 Testing examples for package 'MASS'
23   comparing 'MASS-Ex.Rout' to 'MASS-Ex.Rout.save' ... OK
24 Testing examples for package 'lattice'
25 Testing examples for package 'Matrix'
26 Testing examples for package 'nlme'
27 Testing examples for package 'survival'
28   comparing 'survival-Ex.Rout' to 'survival-Ex.Rout.save' ... OK
29 Testing examples for package 'boot'
30   comparing 'boot-Ex.Rout' to 'boot-Ex.Rout.save' ... OK
31 Testing examples for package 'cluster'
32 Testing examples for package 'codetools'
33 Testing examples for package 'foreign'
34 Testing examples for package 'KernSmooth'
35 Testing examples for package 'rpart'
36   comparing 'rpart-Ex.Rout' to 'rpart-Ex.Rout.save' ... OK
37 Testing examples for package 'class'
38 Testing examples for package 'nnet'
39 Testing examples for package 'spatial'
40   comparing 'spatial-Ex.Rout' to 'spatial-Ex.Rout.save' ... OK
41 Testing examples for package 'mgcv'
42
43
44 Test suite result: PASS
```

The final line of the above output displays the status of running the above tests. **PASS** indicates a successful running of the tests, a **FAIL** would indicate that an error was detected during the running of the tests.

There may be some tests where the result of performing a `diff` on two files that were being compared demonstrate a content difference that may or may not be relevant and may be dependent upon locale settings. Any such differences displayed in the above output should be reviewed in detail to determine their relevance to the Operational Qualification of this R installation.

R Recommended Package Operational Qualification - Package Vignettes (OQ)

The following is the output of `testInstalledPackages(outDir = "IQ-OQ-TestOutput", scope = "recommended", types = "vignettes", errorsAreFatal = FALSE)`, which runs a series of operational tests of the R Recommended package vignette code examples:

```
1
2 R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
3 Copyright (C) 2026 The R Foundation for Statistical Computing
4 Platform: x86_64-w64-mingw32/x64
5
6 R is free software and comes with ABSOLUTELY NO WARRANTY.
7 You are welcome to redistribute it under certain conditions.
8 Type 'license()' or 'licence()' for distribution details.
9
10 Natural language support but running in an English locale
11
12 R is a collaborative project with many contributors.
13 Type 'contributors()' for more information and
14 'citation()' on how to cite R or R packages in publications.
15
16 Type 'demo()' for some demos, 'help()' for on-line help, or
17 'help.start()' for an HTML browser interface to help.
18 Type 'q()' to quit R.
19
20 >
21 > options(echo = FALSE)
22 Running vignettes for package 'lattice'
23 Running 'grid.Rnw'
24 Running vignettes for package 'Matrix'
25 Running 'Comparisons.Rnw'
26 Running 'Design-issues.Rnw'
27 Running 'Intro2Matrix.Rnw'
28 Running 'Introduction.Rnw'
29 Running 'sparseModels.Rnw'
30 Running vignettes for package 'survival'
31 Running 'adjcurve.Rnw'
32 Running 'approximate.Rnw'
33 Running 'compete.Rnw'
34 Running 'concordance.Rnw'
35 Running 'matrix.Rnw'
36 Running 'methods.Rnw'
37 Running 'multi.Rnw'
38 Running 'other.Rnw'
39 Running 'population.Rnw'
40 Running 'redistribute.Rnw'
41 Running 'splines.Rnw'
42 Running 'survival.Rnw'
43 Running 'tiedtimes.Rnw'
44 Running 'timedep.Rnw'
45 Running 'validate.Rnw'
46 Running vignettes for package 'rpart'
47 Running 'longintro.Rnw'
48 Running 'usercode.Rnw'
49
50
51 Test suite result: PASS
```

The final line of the above output displays the status of running the above tests. **PASS** indicates a successful running of the tests, a **FAIL** would indicate that an error was detected during the running of the tests.

There may be some tests where the result of performing a `diff` on two files that were being compared demonstrate a content difference that may or may not be relevant and may be dependent upon locale settings. Any such differences displayed in the above output should be reviewed in detail to determine their relevance to the Operational Qualification of this R installation.

R Base Package Operational Qualification - Package Tests (OQ)

The following is the output of `testInstalledPackages(outDir = "IQ-OQ-TestOutput", scope = "base", types = "tests", errorsAreFatal = FALSE)`, which runs a series of operational tests of the R Base package code tests:

```
1 R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
2 Copyright (C) 2026 The R Foundation for Statistical Computing
3 Platform: x86_64-w64-mingw32/x64
4
5
6 R is free software and comes with ABSOLUTELY NO WARRANTY.
7 You are welcome to redistribute it under certain conditions.
8 Type 'license()' or 'licence()' for distribution details.
9
10 Natural language support but running in an English locale
11
12 R is a collaborative project with many contributors.
13 Type 'contributors()' for more information and
14 'citation()' on how to cite R or R packages in publications.
15
16 Type 'demo()' for some demos, 'help()' for on-line help, or
17 'help.start()' for an HTML browser interface to help.
18 Type 'q()' to quit R.
19
20 >
21 > options(echo = FALSE)
22 Running specific tests for package 'tools'
23   Running 'bibtools.R'
24   comparing 'bibtools.Rout' to 'bibtools.Rout.save' ... OK
25   Running 'hashes.R'
26   Running 'QC.R'
27   Running 'Rd.R'
28   Running 'Rd2HTML.R'
29   Running 'Rd2pdf.R'
30   Running 'S3.R'
31   Running 'undoc.R'
32 Running specific tests for package 'utils'
33   Running 'charclass.R'
34   Running 'completion.R'
35   Running 'relist.R'
36   Running 'Stangle-tst.R'
37   Running 'Sweave-tst.R'
38   Running 'SweaveGetSourceName-tst.R'
39   Running 'tar.R'
40 Running specific tests for package 'grDevices'
41   Running 'convertColor-tests.R'
42   Running 'encodings.R'
43   comparing 'encodings.Rout' to 'encodings.Rout.save' ... OK
44   Running 'encodings2.R'
45   comparing 'encodings2.Rout' to 'encodings2.Rout.save' ... OK
46   Running 'encodings3.R'
47   comparing 'encodings3.Rout' to 'encodings3.Rout.save' ... OK
48   Running 'grDev-tsts.R'
49   Running 'palettes-tests.R'
50   Running 'ps-tests.R'
51   comparing 'ps-tests.Rout' to 'ps-tests.Rout.save' ... OK
52   Running 'saved-recordPlot.R'
53   Running 'urw-fonts.R'
54   Running 'xyTable.R'
55   comparing 'xyTable.Rout' to 'xyTable.Rout.save' ... OK
56   Running 'zzcheck-encodings.R'
57 Running specific tests for package 'stats'
58   Running 'arimaML.R'
59   Running 'bandwidth.R'
60   comparing 'bandwidth.Rout' to 'bandwidth.Rout.save' ... OK
61   Running 'cmdscale.R'
62   Running 'density_chk.R'
63   Running 'dpq-extra.R'
64   Running 'drop1-polr.R'
65   Running 'factanal-tst.R'
66   Running 'glm-etc.R'
67   Running 'glm.R'
68   comparing 'glm.Rout' to 'glm.Rout.save' ... OK
69   Running 'ig_glm.R'
70   Running 'ks-test.R'
71   comparing 'ks-test.Rout' to 'ks-test.Rout.save' ... OK
72   Running 'loglin.R'
73   comparing 'loglin.Rout' to 'loglin.Rout.save' ... OK
74   Running 'nafns.R'
```

```

75 Running 'nlm.R'
76 Running 'nls.R'
77 comparing 'nls.Rout' to 'nls.Rout.save' ... OK
78 Running 'NLSstClosest.R'
79 Running 'offsets.R'
80 Running 'ppr.R'
81 Running 'psmirnov.R'
82 comparing 'psmirnov.Rout' to 'psmirnov.Rout.save' ... OK
83 Running 'simulate.R'
84 comparing 'simulate.Rout' to 'simulate.Rout.save' ... OK
85 Running 'smooth.spline.R'
86 Running 'table-margins.R'
87 Running 'ts-tests.R'
88 Running specific tests for package 'methods'
89 Running 'basicRefClass.R'
90 Running 'duplicateClass.R'
91 Running 'envRefClass.R'
92 Running 'fieldAssignments.R'
93 Running 'mixinInitialize.R'
94 Running 'namesAndSlots.R'
95 Running 'nextWithDots.R'
96 Running 'refClassExample.R'
97 Running 'S3.R'
98 Running 'testConditionalIs.R'
99 Running 'testGroupGeneric.R'
100 Running 'testIs.R'
101 Running specific tests for package 'grid'
102 Running 'bugs.R'
103 Running 'clippaths.R'
104 Running 'compositing.R'
105 Running 'coords.R'
106 Running 'glyphs.R'
107 Running 'grep.R'
108 comparing 'grep.Rout' to 'grep.Rout.save' ... OK
109 Running 'groups.R'
110 Running 'masks.R'
111 Running 'nesting.R'
112 Running 'paths.R'
113 Running 'patterns.R'
114 Running 'reg.R'
115 Running 'testIs.R'
116 comparing 'testIs.Rout' to 'testIs.Rout.save' ... OK
117 Running 'units.R'
118 Running specific tests for package 'splines'
119 Running 'sparse-tst.R'
120 Running 'spline-tst.R'
121 Running specific tests for package 'stats4'
122 Running 'confint.R'
123 Running specific tests for package 'compiler'
124 Running 'assign.R'
125 Running 'basics.R'
126 Running 'const.R'
127 Running 'curexpr.R'
128 Running 'envir.R'
129 Running 'jit.R'
130 Running 'loop.R'
131 Running 'srcref.R'
132 Running 'switch.R'
133 Running 'vischk.R'
134 Running specific tests for package 'parallel'
135 Running 'Master.R'
136 Running 'RSeed.R'
137
138
139 Test suite result: PASS

```

The final line of the above output displays the status of running the above tests. PASS indicates a successful running of the tests, a FAIL would indicate that an error was detected during the running of the tests.

There may be some tests where the result of performing a diff on two files that were being compared demonstrate a content difference that may or may not be relevant and may be dependent upon locale settings. Any such differences displayed in the above output should be reviewed in detail to determine their relevance to the Operational Qualification of this R installation.

R Recommended Package Operational Qualification - Package Tests (OQ)

The following is the output of `testInstalledPackages(outDir = "IQ-OQ-TestOutput", scope = "recommended", types = "tests", errorsAreFatal = FALSE)`, which runs a series of operational tests of the R Recommended package code tests:

```
1 R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
2 Copyright (C) 2026 The R Foundation for Statistical Computing
3 Platform: x86_64-w64-mingw32/x64
4
5
6 R is free software and comes with ABSOLUTELY NO WARRANTY.
7 You are welcome to redistribute it under certain conditions.
8 Type 'license()' or 'licence()' for distribution details.
9
10 Natural language support but running in an English locale
11
12 R is a collaborative project with many contributors.
13 Type 'contributors()' for more information and
14 'citation()' on how to cite R or R packages in publications.
15
16 Type 'demo()' for some demos, 'help()' for on-line help, or
17 'help.start()' for an HTML browser interface to help.
18 Type 'q()' to quit R.
19
20 >
21 > options(echo = FALSE)
22 Running specific tests for package 'MASS'
23   Running 'confint.R'
24   Running 'cov.mcd.R'
25   Running 'fitdistr.R'
26   comparing 'fitdistr.Rout' to 'fitdistr.Rout.save' ... OK
27   Running 'glm.nb.R'
28   Running 'glmmPQL.R'
29   Running 'hubers.R'
30   Running 'lme.R'
31   Running 'loglm.R'
32   Running 'polr.R'
33   Running 'profile.R'
34   Running 'regression.R'
35   comparing 'regression.Rout' to 'regression.Rout.save' ... OK
36   Running 'rlm.R'
37   Running 'scripts.R'
38 Running specific tests for package 'lattice'
39   Running 'auto-key.R'
40   Running 'barchart-width.R'
41   Running 'call.R'
42   Running 'colorkey-title.R'
43   Running 'dataframe-methods.R'
44   Running 'dates.R'
45   Running 'dotplotscoping.R'
46   Running 'fontsize.R'
47   Running 'levelplot.R'
48   Running 'MASSch04.R'
49   Running 'scales.R'
50   Running 'shade-wireframe.R'
51   Running 'summary.R'
52   Running 'temp.R'
53   Running 'test.R'
54   Running 'wireframe.R'
55 Running specific tests for package 'Matrix'
56   Running 'abIndex-tsts.R'
57   Running 'base-matrix-fun.R'
58   Running 'bind.R'
59   comparing 'bind.Rout' to 'bind.Rout.save' ... OK
60   Running 'Class+Meth.R'
61   Running 'dg_Matrix.R'
62   Running 'dpo-test.R'
63   Running 'dtpMatrix.R'
64   Running 'factorizing.R'
65   Running 'group-methods.R'
66   Running 'indexing.R'
67   comparing 'indexing.Rout' to 'indexing.Rout.save' ... OK
68   Running 'matprod.R'
69   Running 'matr-exp.R'
70   Running 'other-pkgs.R'
71   Running 'packed-unpacked.R'
72   Running 'Simple.R'
73   Running 'spModel.matrix.R'
74   Running 'symmDN.R'
```

```

75 Running 'validObj.R'
76 Running 'write-read.R'
77 Running specific tests for package 'nlme'
78 Running 'anova.gls.R'
79 Running 'augPred_lab.R'
80 Running 'augPredmissing.R'
81 Running 'coef.R'
82 comparing 'coef.Rout' to 'coef.Rout.save' ... OK
83 Running 'contrMat.R'
84 Running 'corMatrix.R'
85 Running 'corStruct.R'
86 Running 'data.frame.R'
87 Running 'deparse.R'
88 Running 'deviance.R'
89 Running 'fitted.R'
90 Running 'getData.R'
91 Running 'getVarCov.R'
92 Running 'glms.R'
93 Running 'gnls-ch8.R'
94 Running 'lme.R'
95 comparing 'lme.Rout' to 'lme.Rout.save' ... OK
96 Running 'lmList.R'
97 Running 'missing.R'
98 comparing 'missing.Rout' to 'missing.Rout.save' ... OK
99 Running 'nlme.R'
100 Running 'nlme2.R'
101 Running 'predict.lme.R'
102 Running 'scoping.R'
103 Running 'sigma-fixed-etc.R'
104 Running 'updateLme.R'
105 Running 'varConstProp.R'
106 Running 'varFixed.R'
107 Running 'varIdent.R'
108 Running specific tests for package 'survival'
109 Running 'aareg.R'
110 comparing 'aareg.Rout' to 'aareg.Rout.save' ... OK
111 Running 'anova.R'
112 comparing 'anova.Rout' to 'anova.Rout.save' ... OK
113 Running 'bladder.R'
114 comparing 'bladder.Rout' to 'bladder.Rout.save' ... OK
115 Running 'book1.R'
116 comparing 'book1.Rout' to 'book1.Rout.save' ... OK
117 Running 'book2.R'
118 comparing 'book2.Rout' to 'book2.Rout.save' ... OK
119 Running 'book3.R'
120 comparing 'book3.Rout' to 'book3.Rout.save' ... OK
121 Running 'book4.R'
122 comparing 'book4.Rout' to 'book4.Rout.save' ... OK
123 Running 'book5.R'
124 comparing 'book5.Rout' to 'book5.Rout.save' ... OK
125 Running 'book6.R'
126 comparing 'book6.Rout' to 'book6.Rout.save' ... OK
127 Running 'book7.R'
128 comparing 'book7.Rout' to 'book7.Rout.save' ... OK
129 Running 'brier.R'
130 comparing 'brier.Rout' to 'brier.Rout.save' ... OK
131 Running 'cancer.R'
132 comparing 'cancer.Rout' to 'cancer.Rout.save' ... OK
133 Running 'checkSurv2.R'
134 comparing 'checkSurv2.Rout' to 'checkSurv2.Rout.save' ... OK
135 Running 'clogit.R'
136 comparing 'clogit.Rout' to 'clogit.Rout.save' ... OK
137 Running 'concordance.R'
138 comparing 'concordance.Rout' to 'concordance.Rout.save' ... OK
139 Running 'concordance2.R'
140 comparing 'concordance2.Rout' to 'concordance2.Rout.save' ... OK
141 Running 'concordance3.R'
142 comparing 'concordance3.Rout' to 'concordance3.Rout.save' ... OK
143 Running 'counting.R'
144 comparing 'counting.Rout' to 'counting.Rout.save' ... OK
145 Running 'coxsurv.R'
146 comparing 'coxsurv.Rout' to 'coxsurv.Rout.save' ... OK
147 Running 'coxsurv2.R'
148 comparing 'coxsurv2.Rout' to 'coxsurv2.Rout.save' ... OK
149 Running 'coxsurv3.R'
150 comparing 'coxsurv3.Rout' to 'coxsurv3.Rout.save' ... OK
151 Running 'coxsurv4.R'
152 comparing 'coxsurv4.Rout' to 'coxsurv4.Rout.save' ... OK
153 Running 'coxsurv5.R'
154 comparing 'coxsurv5.Rout' to 'coxsurv5.Rout.save' ... OK
155 Running 'coxsurv6.R'

```

```

156 comparing 'coxsurv6.Rout' to 'coxsurv6.Rout.save' ... OK
157 Running 'detail.R'
158 comparing 'detail.Rout' to 'detail.Rout.save' ... OK
159 Running 'diffptest.R'
160 comparing 'diffptest.Rout' to 'diffptest.Rout.save' ... OK
161 Running 'doaml.R'
162 comparing 'doaml.Rout' to 'doaml.Rout.save' ... OK
163 Running 'doublecolon.R'
164 comparing 'doublecolon.Rout' to 'doublecolon.Rout.save' ... OK
165 Running 'doweight.R'
166 comparing 'doweight.Rout' to 'doweight.Rout.save' ... OK
167 Running 'dropspecial.R'
168 comparing 'dropspecial.Rout' to 'dropspecial.Rout.save' ... OK
169 Running 'ekm.R'
170 comparing 'ekm.Rout' to 'ekm.Rout.save' ... OK
171 Running 'expected.R'
172 comparing 'expected.Rout' to 'expected.Rout.save' ... OK
173 Running 'expected2.R'
174 comparing 'expected2.Rout' to 'expected2.Rout.save' ... OK
175 Running 'factor.R'
176 comparing 'factor.Rout' to 'factor.Rout.save' ... OK
177 Running 'factor2.R'
178 comparing 'factor2.Rout' to 'factor2.Rout.save' ... OK
179 Running 'finegray.R'
180 comparing 'finegray.Rout' to 'finegray.Rout.save' ... OK
181 Running 'fr_cancer.R'
182 comparing 'fr_cancer.Rout' to 'fr_cancer.Rout.save' ... OK
183 Running 'fr_kidney.R'
184 comparing 'fr_kidney.Rout' to 'fr_kidney.Rout.save' ... OK
185 Running 'fr_lung.R'
186 comparing 'fr_lung.Rout' to 'fr_lung.Rout.save' ... OK
187 Running 'fr_ovarian.R'
188 comparing 'fr_ovarian.Rout' to 'fr_ovarian.Rout.save' ... OK
189 Running 'fr_rat1.R'
190 comparing 'fr_rat1.Rout' to 'fr_rat1.Rout.save' ... OK
191 Running 'fr_resid.R'
192 comparing 'fr_resid.Rout' to 'fr_resid.Rout.save' ... OK
193 Running 'fr_simple.R'
194 comparing 'fr_simple.Rout' to 'fr_simple.Rout.save' ... OK
195 Running 'frailty.R'
196 comparing 'frailty.Rout' to 'frailty.Rout.save' ... OK
197 Running 'frank.R'
198 comparing 'frank.Rout' to 'frank.Rout.save' ... OK
199 Running 'infcox.R'
200 comparing 'infcox.Rout' to 'infcox.Rout.save' ... OK
201 Running 'jasa.R'
202 comparing 'jasa.Rout' to 'jasa.Rout.save' ... OK
203 Running 'model.matrix.R'
204 comparing 'model.matrix.Rout' to 'model.matrix.Rout.save' ... OK
205 Running 'mstate.R'
206 comparing 'mstate.Rout' to 'mstate.Rout.save' ... OK
207 Running 'mstate2.R'
208 comparing 'mstate2.Rout' to 'mstate2.Rout.save' ... OK
209 Running 'mstrata.R'
210 comparing 'mstrata.Rout' to 'mstrata.Rout.save' ... OK
211 Running 'multi2.R'
212 comparing 'multi2.Rout' to 'multi2.Rout.save' ... OK
213 Running 'multi3.R'
214 comparing 'multi3.Rout' to 'multi3.Rout.save' ... OK
215 Running 'multistate.R'
216 comparing 'multistate.Rout' to 'multistate.Rout.save' ... OK
217 Running 'multistrata.R'
218 Running 'neardate.R'
219 comparing 'neardate.Rout' to 'neardate.Rout.save' ... OK
220 Running 'nested.R'
221 comparing 'nested.Rout' to 'nested.Rout.save' ... OK
222 Running 'ovarian.R'
223 comparing 'ovarian.Rout' to 'ovarian.Rout.save' ... OK
224 Running 'overlap.R'
225 comparing 'overlap.Rout' to 'overlap.Rout.save' ... OK
226 Running 'prednew.R'
227 comparing 'prednew.Rout' to 'prednew.Rout.save' ... OK
228 Running 'predsurv.R'
229 comparing 'predsurv.Rout' to 'predsurv.Rout.save' ... OK
230 Running 'pseudo.R'
231 comparing 'pseudo.Rout' to 'pseudo.Rout.save' ... OK
232 Running 'pspline.R'
233 comparing 'pspline.Rout' to 'pspline.Rout.save' ... OK
234 Running 'pyear.R'
235 comparing 'pyear.Rout' to 'pyear.Rout.save' ... OK
236 Running 'quantile.R'

```

```

237 comparing 'quantile.Rout' to 'quantile.Rout.save' ... OK
238 Running 'r_lung.R'
239 comparing 'r_lung.Rout' to 'r_lung.Rout.save' ... OK
240 Running 'r_resid.R'
241 comparing 'r_resid.Rout' to 'r_resid.Rout.save' ... OK
242 Running 'r_sas.R'
243 comparing 'r_sas.Rout' to 'r_sas.Rout.save' ... OK
244 Running 'r_scale.R'
245 comparing 'r_scale.Rout' to 'r_scale.Rout.save' ... OK
246 Running 'r_stanford.R'
247 comparing 'r_stanford.Rout' to 'r_stanford.Rout.save' ... OK
248 Running 'r_strata.R'
249 comparing 'r_strata.Rout' to 'r_strata.Rout.save' ... OK
250 Running 'r_tdist.R'
251 comparing 'r_tdist.Rout' to 'r_tdist.Rout.save' ... OK
252 Running 'r_user.R'
253 comparing 'r_user.Rout' to 'r_user.Rout.save' ... OK
254 Running 'ratetable.R'
255 comparing 'ratetable.Rout' to 'ratetable.Rout.save' ... OK
256 Running 'residms.R'
257 comparing 'residms.Rout' to 'residms.Rout.save' ... OK
258 Running 'residsf.R'
259 comparing 'residsf.Rout' to 'residsf.Rout.save' ... OK
260 Running 'royston.R'
261 comparing 'royston.Rout' to 'royston.Rout.save' ... OK
262 Running 'rttright.R'
263 comparing 'rttright.Rout' to 'rttright.Rout.save' ... OK
264 Running 'singtest.R'
265 comparing 'singtest.Rout' to 'singtest.Rout.save' ... OK
266 Running 'strata2.R'
267 comparing 'strata2.Rout' to 'strata2.Rout.save' ... OK
268 Running 'stratatest.R'
269 comparing 'stratatest.Rout' to 'stratatest.Rout.save' ... OK
270 Running 'summary_survfit.R'
271 comparing 'summary_survfit.Rout' to 'summary_survfit.Rout.save' ... OK
272 Running 'summarydf.R'
273 Running 'surv.R'
274 comparing 'surv.Rout' to 'surv.Rout.save' ... OK
275 Running 'survcheck.R'
276 comparing 'survcheck.Rout' to 'survcheck.Rout.save' ... OK
277 Running 'survexpm.R'
278 Running 'survfit1.R'
279 comparing 'survfit1.Rout' to 'survfit1.Rout.save' ... OK
280 Running 'survfit2.R'
281 comparing 'survfit2.Rout' to 'survfit2.Rout.save' ... OK
282 Running 'survreg1.R'
283 comparing 'survreg1.Rout' to 'survreg1.Rout.save' ... OK
284 Running 'survreg2.R'
285 comparing 'survreg2.Rout' to 'survreg2.Rout.save' ... OK
286 Running 'survSplit.R'
287 comparing 'survSplit.Rout' to 'survSplit.Rout.save' ... OK
288 Running 'survtest.R'
289 comparing 'survtest.Rout' to 'survtest.Rout.save' ... OK
290 Running 'testci.R'
291 comparing 'testci.Rout' to 'testci.Rout.save' ... OK
292 Running 'testci2.R'
293 comparing 'testci2.Rout' to 'testci2.Rout.save' ... OK
294 Running 'testnull.R'
295 comparing 'testnull.Rout' to 'testnull.Rout.save' ... OK
296 Running 'testreg.R'
297 comparing 'testreg.Rout' to 'testreg.Rout.save' ... OK
298 Running 'tiedtime.R'
299 comparing 'tiedtime.Rout' to 'tiedtime.Rout.save' ... OK
300 Running 'tmerge.R'
301 comparing 'tmerge.Rout' to 'tmerge.Rout.save' ... OK
302 Running 'tmerge2.R'
303 comparing 'tmerge2.Rout' to 'tmerge2.Rout.save' ... OK
304 Running 'tmerge3.R'
305 comparing 'tmerge3.Rout' to 'tmerge3.Rout.save' ... OK
306 Running 'tmerge4.R'
307 comparing 'tmerge4.Rout' to 'tmerge4.Rout.save' ... OK
308 Running 'tt.R'
309 comparing 'tt.Rout' to 'tt.Rout.save' ... OK
310 Running 'tt2.R'
311 comparing 'tt2.Rout' to 'tt2.Rout.save' ... OK
312 Running 'turnbull.R'
313 comparing 'turnbull.Rout' to 'turnbull.Rout.save' ... OK
314 Running 'update.R'
315 comparing 'update.Rout' to 'update.Rout.save' ... OK
316 Running 'yates0.R'
317 comparing 'yates0.Rout' to 'yates0.Rout.save' ... OK

```

```

318 Running 'yates1.R'
319 comparing 'yates1.Rout' to 'yates1.Rout.save' ... OK
320 Running 'yates2.R'
321 Running 'zph.R'
322 comparing 'zph.Rout' to 'zph.Rout.save' ... OK
323 Running specific tests for package 'boot'
324 Running 'parallel-censboot.R'
325 Running specific tests for package 'cluster'
326 Running 'agnes-ex.R'
327 comparing 'agnes-ex.Rout' to 'agnes-ex.Rout.save' ... OK
328 Running 'clara-ex.R'
329 comparing 'clara-ex.Rout' to 'clara-ex.Rout.save' ... OK
330 Running 'clara-gower.R'
331 Running 'clara-NAs.R'
332 comparing 'clara-NAs.Rout' to 'clara-NAs.Rout.save' ... OK
333 Running 'clara.R'
334 comparing 'clara.Rout' to 'clara.Rout.save' ... OK
335 Running 'clusplot-out.R'
336 comparing 'clusplot-out.Rout' to 'clusplot-out.Rout.save' ... OK
337 Running 'daisy-ex.R'
338 comparing 'daisy-ex.Rout' to 'daisy-ex.Rout.save' ... OK
339 Running 'diana-boots.R'
340 Running 'diana-ex.R'
341 comparing 'diana-ex.Rout' to 'diana-ex.Rout.save' ... OK
342 Running 'ellipsoid-ex.R'
343 comparing 'ellipsoid-ex.Rout' to 'ellipsoid-ex.Rout.save' ... OK
344 Running 'fanny-ex.R'
345 comparing 'fanny-ex.Rout' to 'fanny-ex.Rout.save' ... OK
346 Running 'mona.R'
347 comparing 'mona.Rout' to 'mona.Rout.save' ... OK
348 Running 'pam.R'
349 comparing 'pam.Rout' to 'pam.Rout.save' ... OK
350 Running 'silhouette-default.R'
351 comparing 'silhouette-default.Rout' to 'silhouette-default.Rout.save' ... OK
352 Running 'sweep-ex.R'
353 Running specific tests for package 'codetools'
354 Running 'tests.R'
355 Running specific tests for package 'foreign'
356 Running 'arff.R'
357 comparing 'arff.Rout' to 'arff.Rout.save' ... OK
358 Running 'download.R'
359 Running 'minitab.R'
360 comparing 'minitab.Rout' to 'minitab.Rout.save' ... OK
361 Running 'mval_bug.R'
362 comparing 'mval_bug.Rout' to 'mval_bug.Rout.save' ... OK
363 Running 'octave.R'
364 comparing 'octave.Rout' to 'octave.Rout.save' ... OK
365 Running 'S3.R'
366 comparing 'S3.Rout' to 'S3.Rout.save' ... OK
367 Running 'sas.R'
368 Running 'spss.R'
369 comparing 'spss.Rout' to 'spss.Rout.save' ... 356c356
370 < $ factor_s_duplicated : Factor w/ 5 levels "A","Ãd","A_duplicated_b",...: 1 5 2 NA NA
371 ---
372 > $ factor_s_duplicated : Factor w/ 5 levels "A","A_duplicated_b",...: 1 5 4 NA NA
373 461c461
374 < $ string_500 : Factor w/ 4 levels "
    | __truncated__,...: 2 1 4 1 3
375 ---
376 > $ string_500 : Factor w/ 4 levels "
    | __truncated__,...: 2 1 3 1 4
377 465c465
378 < $ factor_s_duplicated : Factor w/ 5 levels "A","Ãd","A_duplicated_b",...: 1 5 2 NA NA
379 ---
380 > $ factor_s_duplicated : Factor w/ 5 levels "A","A_duplicated_b",...: 1 5 4 NA NA
381 Running 'stata.R'
382 comparing 'stata.Rout' to 'stata.Rout.save' ... OK
383 Running 'testEmpty.R'
384 comparing 'testEmpty.Rout' to 'testEmpty.Rout.save' ... OK
385 Running 'writeForeignSPSS.R'
386 comparing 'writeForeignSPSS.Rout' to 'writeForeignSPSS.Rout.save' ... OK
387 Running 'xport.R'
388 comparing 'xport.Rout' to 'xport.Rout.save' ... OK
389 Running specific tests for package 'KernSmooth'
390 Running 'bkfe.R'
391 Running 'locpoly.R'
392 Running specific tests for package 'rpart'
393 Running 'backticks.R'
394 comparing 'backticks.Rout' to 'backticks.Rout.save' ... OK

```

```

395 Running 'cost.R'
396 comparing 'cost.Rout' to 'cost.Rout.save' ... OK
397 Running 'cptest.R'
398 comparing 'cptest.Rout' to 'cptest.Rout.save' ... OK
399 Running 'minus_in_formula.R'
400 comparing 'minus_in_formula.Rout' to 'minus_in_formula.Rout.save' ... OK
401 Running 'priors.R'
402 comparing 'priors.Rout' to 'priors.Rout.save' ... OK
403 Running 'rescale.R'
404 comparing 'rescale.Rout' to 'rescale.Rout.save' ... OK
405 Running 'testall.R'
406 comparing 'testall.Rout' to 'testall.Rout.save' ... OK
407 Running 'treble.R'
408 comparing 'treble.Rout' to 'treble.Rout.save' ... OK
409 Running 'treble2.R'
410 comparing 'treble2.Rout' to 'treble2.Rout.save' ... OK
411 Running 'treble3.R'
412 comparing 'treble3.Rout' to 'treble3.Rout.save' ... OK
413 Running 'treble4.R'
414 comparing 'treble4.Rout' to 'treble4.Rout.save' ... OK
415 Running 'usersplits.R'
416 comparing 'usersplits.Rout' to 'usersplits.Rout.save' ... OK
417 Running 'xpred1.R'
418 comparing 'xpred1.Rout' to 'xpred1.Rout.save' ... OK
419 Running 'xpred2.R'
420 comparing 'xpred2.Rout' to 'xpred2.Rout.save' ... OK
421 Running specific tests for package 'spatial'
422
423
424 Test suite result: PASS

```

The final line of the above output displays the status of running the above tests. **PASS** indicates a successful running of the tests, a **FAIL** would indicate that an error was detected during the running of the tests.

There may be some tests where the result of performing a **diff** on two files that were being compared demonstrate a content difference that may or may not be relevant and may be dependent upon locale settings. Any such differences displayed in the above output should be reviewed in detail to determine their relevance to the Operational Qualification of this R installation.

Summary of Findings

The following table presents the results of the various tests performed in the prior sections.

The column labeled **System Results** is an indication that the individual test batch file was able to be executed (**PASS**) or that there may have been a system level failure (**FAIL**) in the execution of the program.

The column labeled **Test Results** is an indication that the test suites themselves either passed (**PASS**) or failed (**FAIL**) and should be consistent with the final line output for each section of tests. As noted previously, there may be some tests where the result of performing a **diff** on two files that were being compared demonstrate a content difference that may or may not be relevant. These differences, if present, may or may not be based upon system settings such as locale. Tests using **Sweave** may fail during testing. **Sweave** is typically not needed for use as part of the installed software. Any such differences displayed in the prior sections should be reviewed in detail to determine their relevance to the Operational Qualification of this R installation.

The result for **Installation Qualification** is listed in the **Test Results** column only. The result will be **PASS** if `system("R -e 'q()'")` ran successfully, otherwise the result will be **FAIL**.

Table 1: Summary of Test Suite Results

Test Suite	System Results	Test Results
Installation Qualification	NA	PASS
Core Operational Qualification - System Tests	PASS	PASS
Base Package Operational Qualification - Package Examples	PASS	PASS
Base Package Operational Qualification - Package Vignettes	PASS	PASS
Recommended Package Operational Qualification - Package Examples	PASS	PASS
Recommended Package Operational Qualification - Package Vignettes	PASS	PASS
Base Package Operational Qualification - Package Tests	PASS	PASS
Recommended Package Operational Qualification - Package Tests	PASS	PASS